

Section 14 Solution

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2 Exercises

2.1 Simple General Equilibrium Example

Consider an exchange economy in which each agent's utility function is: $u_i(x_1^i, x_2^i) = x_1^i * x_2^i$ for $i = A, B$. Initial endowments are $\omega^A = (2, 8)$ and $\omega^B = (8, 2)$. Find the competitive (Walrasian) equilibrium.

1. Write and solve the utility maximization problem for agent A.

$$\max_{x_1^A, x_2^A} x_1^A x_2^A \quad \text{s.t.} \quad p_1 x_1^A + p_2 x_2^A \leq p_1 w_1^A + p_2 w_2^A$$

So the Lagrangian function can be written as

$$\mathcal{L} = x_1^A x_2^A - \lambda (p_1 x_1^A + p_2 x_2^A - p_1 w_1^A - p_2 w_2^A)$$

FOC:

$$\begin{aligned} x_2^A - \lambda p_1 &= 0 \\ x_1^A - \lambda p_2 &= 0 \\ \implies \frac{x_2^A}{x_1^A} &= \frac{p_1}{p_2} \end{aligned}$$

Plugging this back into agent 1's budget:

$$\begin{aligned} p_1 x_1^A + p_2 x_1^A \frac{p_1}{p_2} &= p_1 w_1^A + p_2 w_2^A \\ x_1^A &= \frac{p_1 w_1^A + p_2 w_2^A}{2p_1} \\ x_2^A &= \frac{p_1 w_1^A + p_2 w_2^A}{2p_2} \end{aligned}$$

2. Write and solve the utility maximization problem for agent B.

We also obtain the same results:

$$\begin{aligned} x_1^B &= \frac{p_1 w_1^B + p_2 w_2^B}{2p_1} \\ x_2^B &= \frac{p_1 w_1^B + p_2 w_2^B}{2p_2} \end{aligned}$$

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3. Using the above demands, plot the offer curves of each player.

Plugging in our endowments to the above equations, we get

$$x_1^A = 1 + 4\frac{p_2}{p_1}, x_2^A = \frac{p_1}{p_2} + 4, x_1^B = 4 + \frac{p_2}{p_1}, x_2^B = 4\frac{p_1}{p_2} + 1$$

Combining Agent A 's demand curves, we get

$$x_2^A = \frac{4}{x_1^A - 1} + 4$$

Similarly, if we combine Agent B 's demand curves, we get

$$x_2^B = \frac{4}{x_1^B - 4} + 1$$

with $x_1^B = 10 - x_1^A$ and $x_2^B = 10 - x_2^A$, we can solve for $x_1^A = x_2^A = x_1^B = x_2^B = 5$.

4. Plug demands for good 1 into the market clearing condition:

The market clearing condition for good 1 is:

$$x_1^A + x_1^B = w_1^A + w_1^B$$

giving

$$\begin{aligned} \frac{p_1 w_1^A + p_2 w_2^A}{2p_1} + \frac{p_1 w_1^B + p_2 w_2^B}{2p_1} &= w_1^A + w_1^B \\ \frac{p_1 (w_1^A + w_1^B) + p_2 (w_2^A + w_2^B)}{2p_1} &= w_1^A + w_1^B \end{aligned}$$

5. Solve for p_2^*/p_1^* and x^{A*}, x^{B*} . [By a result called *Walras' Law*, we only need to solve the market clearing condition for one of the goods to know that the markets for both goods will be in equilibrium at this price.] Does this allocation look like what we found visually in part 3?

We solve for the prices:

$$\begin{aligned} p_1 (w_1^A + w_1^B) + p_2 (w_2^A + w_2^B) &= 2p_1 (w_1^A + w_1^B) \\ p_2 (w_2^A + w_2^B) &= p_1 (w_1^A + w_1^B) \\ \frac{p_1}{p_2} &= \frac{w_2^A + w_2^B}{w_1^A + w_1^B} \end{aligned}$$

We can plug this price ratio back into the demands to get:

$$\begin{aligned} x_1^A &= \frac{w_1^A}{2} + \frac{w_2^A}{2} \frac{w_1^A + w_1^B}{w_2^A + w_2^B} \\ x_2^B &= \frac{w_1^B}{2} \frac{w_2^A + w_2^B}{w_1^A + w_1^B} + \frac{w_2^B}{2} \end{aligned}$$

and,

$$\begin{aligned} x_1^B &= \frac{w_1^B}{2} + \frac{w_2^B}{2} \frac{w_1^A + w_1^B}{w_2^A + w_2^B} \\ x_2^B &= \frac{w_1^B}{2} \frac{w_2^A + w_2^B}{w_1^A + w_1^B} + \frac{w_2^B}{2} \end{aligned}$$

Plugging in $w_2^A = w_1^B = 8$, and $w_1^A = w_2^B = 2$:

$$x_1^A = \frac{2}{2} + \frac{8(10)}{2(10)} = 5 = x_2^A = x_1^B = x_2^B$$

and the price ration will be:

$$\frac{p_1}{p_2} = \frac{10}{10} = 1$$

So with these endowments, the unique Walrasian equilibrium will be the very center point of the Edgeworth box, which is what we expected from our offer curve plots. The price of the two goods will be equal. See the graphical solution below.

6. Yes - this is a consequence of the First Welfare Theorem. We could test it by seeing if the slopes of each agent's indifference curves at the equilibrium are identical. For other allocations, we would disprove Pareto efficiency by finding another allocation that is weakly preferred by both agents, with one player strictly preferring the new allocation.

2.2 Utility Maximization and Barter (Problem 1, Final, Spring 2008)

Consider consumers with preferences $u(x_1, x_2) = \min(x_1, x_2)$ over the two good x_1 and x_2 .

1. Draw the indifference curves for this utility function and provide an example of goods that could approximately fit this pattern of preferences.

Answer:

The two goods x_1 and x_2 are perfect complements in this example, thus the indifference curves for this utility function take an L-shape, as shown in the graph below. An example of goods could be any pair of goods that are used together in a strict one-to-one ratio like the example of left and right shoes.

2. Consider a standard utility maximization problem. Each consumer solves the problem

$$\begin{aligned} \max_{x_1, x_2} \quad & \min(x_1, x_2) \\ \text{s.t.} \quad & p_1 x_1 + p_2 x_2 \leq M \end{aligned}$$

Solve for $x_1^*(p_1, p_2, M)$ and $x_2^*(p_1, p_2, M)$. Provide the steps for the solution. [Hint: Do not go for straight for the derivatives, help yourself with the figure in point 1]

Answer:

Graphically, one can see that the optimum will be at the kink of the indifference curves, where $x_1^* = x_2^*$. To find the optimum, substitute $x_1^* = x_2^*$ in to the budget constraint to get $p_1 x_1^* + p_2 x_1^* = M$ yielding $x_1^* = x_2^* = M / (p_1 + p_2)$.

3. Compute how the optimal solutions for $x_1^*(p_1, p_2, M)$ and $x_2^*(p_1, p_2, M)$ varies as a function of p_1 and p_2 . Provide economic intuition.

Answer:

The optimal solutions for x_1 and x_2 vary exactly the same way for both prices as shown below by the partial derivatives. $\frac{\partial x_1^*(p_1, p_2, M)}{\partial p_1} = \frac{\partial x_2^*(p_1, p_2, M)}{\partial p_1} = \frac{\partial x_1^*(p_1, p_2, M)}{\partial p_2} = \frac{\partial x_2^*(p_1, p_2, M)}{\partial p_2} = -\frac{M}{(p_1 + p_2)^2} < 0$

4. Now solve a problem similar to the one in point 2, except that each consumer is endowed not with income M , but with initial endowments of the two goods ω_1 and ω_2 . Hence, each consumer solves

$$\begin{aligned} \max_{x_1, x_2} \quad & \min(x_1, x_2) \\ \text{s.t.} \quad & p_1 x_1 + p_2 x_2 \leq p_1 \omega_1 + p_2 \omega_2 \end{aligned}$$

Solve for $x_1^*(p_1, p_2, \omega_1, \omega_2)$ and $x_2^*(p_1, p_2, \omega_1, \omega_2)$

Answer:

Again, one can see that the optimum will be at the kink of the indifference curves, where $x_1^* = x_2^*$. To find the optimum, substitute $x_1^* = x_2^*$ in to the budget constraint to get $p_1 x_1^* + p_2 x_1^* = p_1 \omega_1 + p_2 \omega_2$ yielding $x_1^* = x_2^* = \frac{p_1 \omega_1 + p_2 \omega_2}{p_1 + p_2}$.

5. Compute how the optimal solutions for $x_1^*(p_1, p_2, \omega_1, \omega_2)$ and $x_2^*(p_1, p_2, \omega_1, \omega_2)$ varies as a function of p_1 and p_2 . Compare to your answers in point 3. Provide economic intuition.

Answer:

$$\frac{\partial x_1^*(p_1, p_2, \omega_1, \omega_2)}{\partial p_1} = \frac{\partial x_2^*(p_1, p_2, \omega_1, \omega_2)}{\partial p_1} = \frac{p_2 \cdot (\omega_1 - \omega_2)}{(p_1 + p_2)^2}$$

$$\frac{\partial x_1^*(p_1, p_2, \omega_1, \omega_2)}{\partial p_2} = \frac{\partial x_2^*(p_1, p_2, \omega_1, \omega_2)}{\partial p_2} = \frac{p_1 \cdot (\omega_2 - \omega_1)}{(p_1 + p_2)^2}$$

The intuition is that an p_1 increases makes you become richer if you are endowed with more of x_1 than of x_2 and thus, by the income effect, consume more of both x_1 and x_2 . (Remember that there is no substitution effect for perfect complements.)

6. Now assume that there are two identical consumers, consumers A and B, each with preferences $u(x_1, x_2) = \min(x_1, x_2)$ and initial endowments (ω_1^A, ω_2^A) and (ω_1^B, ω_2^B) . Assume first $(\omega_1^A, \omega_2^A) = (1, 3)$ and $(\omega_1^B, \omega_2^B) = (3, 1)$. Draw the Edgeworth box for this economy and determine the Individually Rational area, that is, the area of points that each individual prefers to the initial allocation.

Answer:

The Edgeworth Box for this economy will be a four-by-four square as $\omega_1 = \omega_1^A + \omega_1^B = 1 + 3$ and $\omega_2 = \omega_2^A + \omega_2^B = 3 + 1$.

Individual Rationality requires that the agent gets at least as much utility from her outcome as she got from her endowment. To identify the Individually Rational area we need to draw the indifference curve of each agent that goes through the endowment point ω . In this example, this will give a two-by-two square in the middle of the Edgeworth Box as the IR area.

More formally, $1 \leq \{x_1^A, x_1^B\} \leq 3$ and $1 \leq \{x_2^A, x_2^B\} \leq 3$.

7. Still on the Edgeworth box, and still assuming $(\omega_1^A, \omega_2^A) = (1, 3)$ and $(\omega_1^B, \omega_2^B) = (3, 1)$, determine the Pareto Set, that is, the set of allocations $(x_1^A, x_2^A), (x_1^B, x_2^B)$ that are feasible and such that there is no other allocation $(\hat{x}_1^A, \hat{x}_2^A), (\hat{x}_1^B, \hat{x}_2^B)$ that *both* consumer would prefer to $(x_1^A, x_2^A), (x_1^B, x_2^B)$ [Notice that this does not restrict the analysis to points in the Individually Rational Area] Argue your logic on this one.

Answer:

Pareto Efficiency requires that at the outcome allocation, there is no “mutually better” outcome possible. Therefore, the Pareto Set will be the diagonal line in this example, that is, where $x_1^A = x_2^A$ and $x_1^B = x_2^B$.

8. Using the answers in points 6 and 7, determine the Contract Curve, that is, the set of points that are possible outcomes of barter starting for $(\omega_1^A, \omega_2^A) = (1, 3)$ and $(\omega_1^B, \omega_2^B) = (3, 1)$.

Answer:

The Contract Curve is the intersection of the Individually Rational area and the Pareto set, thus, it is the diagonal line for the two-by-two square in the middle of the Edgeworth Box.

More formally, $x_1^A = x_2^A, x_1^B = x_2^B, 1 \leq \{x_1^A, x_1^B\} \leq 3$, and $1 \leq \{x_2^A, x_2^B\} \leq 3$.

9. For general (ω_1^A, ω_2^A) and (ω_1^B, ω_2^B) but assuming $\omega_1^A + \omega_1^B = \omega_2^A + \omega_2^B$ (that is, a square Edgeworth box), determine mathematically the expression for the Contract Curve in the barter case for two consumers for preferences $u(x_1, x_2) = \min(x_1, x_2)$.

Answer:

The Contract Curve in general can be described as follows:

$$\begin{aligned} x_1^A &= x_2^A \\ x_1^B &= x_2^B \\ \min\{\omega_1^A, \omega_1^B\} &\leq \{x_1^A, x_1^B\} \leq \max\{\omega_1^A, \omega_1^B\} \\ \min\{\omega_2^A, \omega_2^B\} &\leq \{x_2^A, x_2^B\} \leq \max\{\omega_2^A, \omega_2^B\} \end{aligned}$$

2.3 More general equilibrium

Jon ($i = 1$) and Nick ($i = 2$) are stuck on the island of Evans. There are two goods that are essential to their well-being: shorts ($j = 1$) and T-shirts ($j = 2$). Jon and Nick have different preferences over shorts and T-shirts. Jon's preferences are governed by $U_1(x_1, x_2) = x_1 x_2^2$ and Nick's preferences can be described by $U_2(x_1, x_2) = x_1^2 x_2$.

1. Draw an Edgeworth Box with the following endowments: $\omega_1 = (8, 2)$ and $\omega_2 = (2, 8)$. (5 points)
Figure has been shown below.

figures/fig3_1.png

Strange occurrences happen on Evans and somehow Jon ended up with all the shorts $\omega_1 = (10, 0)$ and Nick ended up with all the T-shirts $\omega_2 = (0, 10)$. Assume that these are the endowments going forward.

1. Compute the marginal rate of substitution for consumers 1 and 2.
Jon's MRS is given by $\frac{\partial U_1 / \partial x_1}{\partial U_1 / \partial x_2} = \frac{x_2^2}{2x_1 x_2} = \frac{x_2}{2x_1}$ and Nick's MRS is given by $\frac{\partial U_2 / \partial x_1}{\partial U_2 / \partial x_2} = \frac{2x_2}{x_1}$. At the initial endowments, these quantities are 0 and ∞ , respectively.
2. Without doing any additional calculations, is the initial endowment a Walrasian equilibrium? How do you know? (5 points)
According to the First Welfare Theorem, the Walrasian equilibrium is Pareto optimal. However, this equilibrium is clearly not Pareto optimal since (a) the MRS's do not equal each other (and the utility function is well-behaved), and (b) everyone getting 5 of each good is a Pareto improvement over 0 utility.
3. Describe the Pareto efficient allocations in this economy (give a mathematical answer and illustrate your answer in the Edgeworth box). (5 points)
Writing things in term of Jon's allocation, the set of Pareto optimal allocations are given by

$$\begin{aligned}\frac{x_2}{2x_1} &= \frac{2(10 - x_2)}{(10 - x_1)} \\ 10x_2 - x_1x_2 &= 40x_1 - 4x_1x_2 \\ (10 + 3x_1)x_2 &= 40x_1 \\ x_2 &= \frac{40x_1}{10 + 3x_1}\end{aligned}$$

4. Write down the budget constraint for each individual. (5 points)
Each individual's budget constraint is:
$$p_1 x_{i1} + p_2 x_{i2} \leq p_1 \omega_{i1} + p_2 \omega_{i2}$$
5. What class of preferences do the agents have? What is particular about the demand derived from these preferences? (5 points)
Each agent has Cobb-Douglas demand, where they spend a fraction of their wealth on the good corresponding to the exponent of said good.
6. Solve for each individual's Marshallian demands as a function of p_1 and p_2 . (5 points)

$$\begin{aligned}
x_{11} &= \frac{1}{3}10 \\
x_{12} &= \frac{2}{3} \frac{10p_1}{p_2} \\
x_{21} &= \frac{2}{3} \frac{10p_2}{p_1} \\
x_{22} &= \frac{1}{3}10
\end{aligned}$$

7. Solve for the market clearing price in the Walrasian equilibrium. Is this price unique? If so, explain why. If not, give an alternative price that would generate the same equilibrium. (5 points)

Clearing the market for good 1:

$$\begin{aligned}
x_{11} + x_{21} &= 10 \\
\frac{10}{3} + \frac{2}{3} \frac{10p_1}{p_2} &= \\
10 + 20 \frac{p_1}{p_2} &= 30 \\
\frac{p_1}{p_2} &= 1
\end{aligned}$$

We remember that we can only solve for relative prices. Therefore, $p_1 = p_2 = 1$ works just as well as $p_2 = p_3 = 17.2$.